

SINTHANAI SELVAN PALANISAMY

Project Research Scientist-I (Biostatistics)

ICMR – National Institute of Epidemiology, Chennai

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PROFESSIONAL SUMMARY

Biostatistician with experience in public health and clinical research, epidemiological studies, and large-scale health surveillance data. Experienced in quantitative research and evidence generation to support epidemiological research, disease surveillance, and public health decision-making.

CORE RESEARCH EXPERTISE

- Statistics/ Biostatistics
- Epidemiology
- Large-Scale Laboratory surveillance data curation
- Infectious Disease Modelling
- Geospatial

PROFESSIONAL EXPERIENCE

Project Research Scientist-I (Biostatistics)

ICMR – National Institute of Epidemiology, Chennai | November 2024 – Present

- Lead statistical analyses for public health and epidemiological research projects.
- Develop Statistical Analysis Plans (SAPs), sampling strategies, and analytical approaches.
- Perform sample size estimation for observational and clinical studies.
- Support protocol development, grant proposals, and manuscript preparation.
- Manage research databases and data quality assurance.
- Conduct geospatial analysis, spatial modelling, and epidemiological analyses.
- Provide biostatistical consultation to multidisciplinary research teams.

Project Technical Officer (Biostatistics)

ICMR – National Institute of Epidemiology | August 2022 – October 2024

- Managed surveillance data from more than 150 VRDL laboratories.
- Developed REDCap eCRFs and data validation rules.
- Prepared publication-quality statistical tables and visualizations.
- Developed SOPs for clinical research data management.
- Conducted statistical training for MPH and M.Sc. students.

Data Analyst

National Health Mission, Tamil Nadu | June 2022 – July 2022

- Worked on Population Health Registry data management and analysis.
- Created dashboards using Google Looker Studio.

EDUCATION

Master of Science in Biostatistics

National Institute of Epidemiology, ICMR School of Public Health | 2020–2022

Bachelor of Science in Statistics

Government Arts and Science College, Kumbakonam | 2017–2020

PUBLICATIONS

- Abdulkader RS, Ramasamy S, Vaman RS, Sapkal GN, Narayan J, Sinthanai SP, et al. Acute encephalitis syndrome and Japanese encephalitis in India: insights from a nationally representative laboratory surveillance network, 2014–2023. *BMC Infectious Diseases* 2025 Dec 8;26(1). DOI: 10.1186/s12879-025-12274-7
- Abdulkader RS, Vijay N, Potdar V, Sinthanai SP, Narayan J, Ramasamy S, et al. Epidemiology of Human Metapneumovirus (HMPV) in India: a cross-sectional study. *The Lancet Regional Health – Southeast Asia* 2025 Sep 17;41:100667. DOI: 10.1016/j.lansea.2025.100667

TECHNICAL SKILLS

Experienced in study design, research protocol development, Statistical Analysis Plan (SAP) development, sampling design, sample size estimation, and power analysis. Proficient in large-scale public health surveillance data curation, clinical trial data management, database management, and REDCap/eCRF development. Skilled in descriptive and inferential statistics, regression analysis, multivariate analysis, time-series analysis, survival analysis, Bayesian analysis and modelling, meta-analysis, and social network analysis. Experienced in statistical modelling, infectious disease modelling, geospatial and spatial modelling, and AI/ML applications. Proficient in data visualization, interactive dashboard development, KPI analytics, web-based data applications, scientific writing, manuscript preparation, and reproducible research.

SOFTWARE SKILLS

Statistical & Programming: R, STATA, SPSS, SAS (Basics), Python, SQL. R ecosystem: Quarto, R Markdown, Shiny. Data capture & management: REDCap. Geospatial: QGIS, GeoDa. Visualization & dashboards: Google Looker Studio. Web technologies: HTML, CSS. Scientific document preparation: LaTeX.

WORKSHOPS & TRAINING

- Sample Size and Power Analysis in Medical Research, NIMHANS, Bengaluru, 2024.
- Workshop on Public Health Legislation, ICMR–NIE, 2025.
- AI in Public Health and Epidemiology Workshop, AI-VRDLN.
- Design and Analysis of Epidemiologic Studies for Causal Inference Workshop, CSMAC, ICMR–NIE, 2024.
- Capacity Building Program on Biostatistical & Clinical Epidemiological Methods, Christian Medical College (CMC), Vellore, 2021.

CONFERENCES & SCIENTIFIC PRESENTATIONS

- Attended India AI Impact Summit 2026.
- Presented a poster at VIROCON 2025, ICMR–National Institute of Virology (NIV): “Outbreak investigations done through the VRDL Network in India, 2014–2024.”

PROFESSIONAL COMPETENCIES

Research methodology, biostatistical consultation, scientific communication, project coordination, multidisciplinary collaboration, research training, manuscript preparation, protocol development, and evidence synthesis.

LANGUAGES

English; Tamil

REFERENCES

Dr Rizwan S A

Scientist – E, ICMR National Institute of Epidemiology, Chennai, Tamil Nadu 600077

Mr. Sabarinathan

Scientist – C, ICMR National Institute of Epidemiology, Chennai, Tamil Nadu 600077